MPG No. 5.3.14 $2'/m$ [Type III, monoclinic] [T tensor]

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} T_x & 0 & T_z \end{bmatrix}$$

$$T_x = \mathbb{T}_x^{(1)}$$
$$T_z = \mathbb{T}_z^{(1)}$$

* Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} T_{xxx} & 0 & T_{xxz} \\ T_{yyx} & 0 & T_{yyz} \\ T_{zzx} & 0 & T_{zzz} \\ 0 & T_{yyz} & 0 \\ T_{xxz} & 0 & T_{zzx} \\ 0 & T_{yyx} & 0 \end{bmatrix}$$

$$T_{xxx} = 3\mathbb{T}_x^{(1)} + 2\mathbb{T}_x^{\alpha(1)}$$
$$T_{xxz} = \mathbb{T}_z^{(1)} - \mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\beta(1)}$$
$$T_{yyx} = \mathbb{T}_x^{(1)} - \mathbb{T}_x^{\alpha(1)} + \mathbb{T}_x^{\beta(1)}$$
$$T_{yyz} = \mathbb{T}_z^{(1)} - \mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\beta(1)}$$
$$T_{zzx} = \mathbb{T}_x^{(1)} - \mathbb{T}_x^{\alpha(1)} - \mathbb{T}_x^{\beta(1)}$$
$$T_{zzz} = 3\mathbb{T}_z^{(1)} + 2\mathbb{T}_z^{\alpha(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & T_{xxz} \\ T_{yyx} & 0 & T_{yyz} \\ T_{zzx} & 0 & 0 \\ 0 & -\frac{T_{yyz}}{2} & 0 \\ -\frac{T_{xxz}}{2} & 0 & -\frac{T_{zzx}}{2} \\ 0 & -\frac{T_{yyx}}{2} & 0 \end{bmatrix}$$

$$T_{xxz} = 2\mathbb{M}_{xy}^{(1)} + 2\mathbb{T}_z^{(2)}$$
$$T_{yyx} = 2\mathbb{M}_{yz}^{(1)} + 2\mathbb{T}_x^{(2)}$$
$$T_{yyz} = -2\mathbb{M}_{xy}^{(1)} + 2\mathbb{T}_z^{(2)}$$
$$T_{zzx} = -2\mathbb{M}_{yz}^{(1)} + 2\mathbb{T}_x^{(2)}$$
$$\frac{T_{yyz}}{2} = -\mathbb{M}_{xy}^{(1)} + \mathbb{T}_z^{(2)}$$
$$\frac{T_{xxz}}{2} = \mathbb{M}_{xy}^{(1)} + \mathbb{T}_z^{(2)}$$
$$\frac{T_{zzx}}{2} = -\mathbb{M}_{yz}^{(1)} + \mathbb{T}_x^{(2)}$$
$$\frac{T_{yyx}}{2} = \mathbb{M}_{yz}^{(1)} + \mathbb{T}_x^{(2)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & T_{yzy} & 0 \\ T_{zxx} & 0 & T_{zxz} \\ 0 & T_{xyy} & 0 \end{bmatrix}$$

$$T_{yzy} = \mathbb{M}_{xy}^{(2)} + \mathbb{T}_z^{(3)}$$
$$T_{zxx} = \mathbb{M}_{xy}^{(2)} - \mathbb{T}_z^{(3)}$$
$$T_{zxz} = \mathbb{M}_{yz}^{(2)} + \mathbb{T}_x^{(3)}$$
$$T_{xyy} = \mathbb{M}_{yz}^{(2)} - \mathbb{T}_x^{(3)}$$

* Rank 4 tensor (sss). * Rank 4 tensor (ssa). * Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa). * Rank 4 tensor (as). * Rank 4 tensor (s). * Rank 4 tensor (a). * Rank 4 tensor (t).